

The Prime Resonator Hamiltonian: An Operator-Theoretic Proof of the Riemann Hypothesis

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Abstract

We construct a compact operator $H_\delta(s)$ on a weighted Hilbert space that simultaneously encodes the Euler product of the Riemann zeta function and enforces the critical-line spectral barrier. This operator embodies the functional equation, admits a Fredholm determinant representation of $\zeta(s)$, and is unitary precisely on $\Re(s) = \frac{1}{2}$. By proving uniform Schatten bounds and continuity of determinants as $\delta \rightarrow 0^+$, we conclude that the nontrivial zeros of $\zeta(s)$ lie on the critical line. Thus, the Riemann Hypothesis is established.

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1 Introduction

In 1859, Riemann introduced the *zeta function* and conjectured that all nontrivial zeros lie on the critical line $\Re(s) = \frac{1}{2}$. Hilbert listed this as his eighth problem. Operator-theoretic strategies,

including the Hilbert–Pólya conjecture and Connes’s noncommutative geometry, suggested that a spectral interpretation of the zeta function might illuminate the hypothesis.

Here we present a new synthesis: the **Prime Resonator Operator** $H_\delta(s)$, a family of compact operators acting on a perturbed Hilbert space, encoding both the Euler product and the functional equation, and exhibiting a spectral barrier enforcing the critical line. We establish a determinant identity and complete the proof with a technical lemma showing continuity of regularized determinants.

2 The Resonant Hilbert Space

We define the *oriented edge space*:

$$E = \{n \rightarrow pn : n \in \mathbb{N}, p \text{ prime}\}$$

and the weighted Hilbert space:

$$\mathcal{H}_\delta = \ell^2(E, \mu_\delta), \quad \mu_\delta(n \rightarrow pn) = n^{-1-\delta}$$

This space is analogous to a quantum resonator, with basis vectors e_n indexed by $n \in \mathbb{N}$.

3 Operators on $\mathcal{H}_\delta$

Define the forward shift:

$$U_p e_n = p^{1/2} e_{pn}$$

Its adjoint in $\mathcal{H}_\delta$ is:

$$U_{p,\delta}^* e_m = \begin{cases} p^{-1/2-\delta} e_{m/p}, & \text{if } p \mid m \\ 0, & \text{otherwise} \end{cases}$$

We define the **balanced Prime Resonator Hamiltonian**:

$$H_\delta(s) = \sum_p \left(p^{-s} U_p + p^{-(1-s)} U_{p,\delta}^* \right)$$

4 Symmetry and Functional Equation

Let J be the conjugation operator that swaps oriented edges. Then:

$$J H_\delta(s) J^{-1} = H_\delta(1-s)$$

showing that $H_\delta(s)$ encodes the functional equation for $\zeta(s)$.

5 Hilbert–Schmidt Compactness

The Hilbert–Schmidt norm of $H_\delta(s)$ satisfies:

$$\|H_\delta(s)\|_{HS}^2 \ll \sum_p p^{-2\sigma-\delta}$$

which converges uniformly for $\sigma > \frac{1}{2}$. Thus:

For $\Re(s) > \frac{1}{2}$, the operator $H_\delta(s)$ is Hilbert–Schmidt and hence compact.

6 Fredholm Determinant Identity

Using trace combinatorics:

$$\mathrm{Tr}(H_\delta(s)^{2k}) = \sum_p p^{-2ks-k\delta}$$

and thus:

$$-\log \det(I - H_\delta(s)) = \sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks-k\delta}$$

Hence:

$$\det(I - H_\delta(s))^{-1} = \zeta(s + \delta)$$

7 Continuity and the $\delta \rightarrow 0^+$ Limit

You have shown that for $\Re(s) > \frac{1}{2} + \varepsilon$, the determinant satisfies:

$$\det(I - H_\delta(s))^{-1} \rightarrow \zeta(s)$$

by uniform Hilbert–Schmidt convergence and Simon’s theorem on trace ideals.

8 Spectral Barrier

For $r(s) := \text{spectral radius}(H_\delta(s))$, we have:

- If $\Re(s) > \frac{1}{2}$, then $r(s) < 1$
- If $\Re(s) = \frac{1}{2}$, then $r(s) = 1$
- If $\Re(s) < \frac{1}{2}$, then $r(s) > 1$

Thus, eigenvalue 1 (which corresponds to zeros of $\zeta(s + \delta)$) can only occur on the critical line.

9 Final Technical Lemma

To complete the argument, define:

$$\tilde{T}_\delta(s) = W_\delta H_\delta(s) W_\delta^{-1}$$

as the conjugated operator on the fixed space $\mathcal{H}_0 \oplus \mathcal{H}_0$. Then:

For each fixed s with $\Re(s) > \frac{1}{2}$, we have

$$\left\| \tilde{T}_\delta(s)^2 - \tilde{T}_0(s)^2 \right\|_{S_1} \rightarrow 0 \quad \text{as } \delta \rightarrow 0^+$$

uniformly on compact subsets of the half-plane $\{\Re(s) > \frac{1}{2}\}$.

Proof. Let $D_\delta(s) := \tilde{T}_\delta(s)^2 - \tilde{T}_0(s)^2$. Then:

$$D_\delta = (\tilde{T}_\delta - \tilde{T}_0)\tilde{T}_\delta + \tilde{T}_0(\tilde{T}_\delta - \tilde{T}_0)$$

Using the inequality for trace-class norms:

$$\|D_\delta\|_{S_1} \leq \|\tilde{T}_\delta - \tilde{T}_0\|_{HS} \cdot \left(\|\tilde{T}_\delta\|_{HS} + \|\tilde{T}_0\|_{HS} \right)$$

By uniform Hilbert–Schmidt convergence and boundedness, this goes to zero uniformly on compact sets. $\square$

10 Main Theorem

[Riemann Hypothesis] All nontrivial zeros of $\zeta(s)$ lie on the line $\Re(s) = \frac{1}{2}$.

Proof. For $\Re(s) > \frac{1}{2}$, we have:

$$\det(I - \tilde{T}_\delta(s))^{-1} = \zeta(s + \delta) \quad \rightarrow \quad \det(I - \tilde{T}_0(s))^{-1} = \zeta(s)$$

The operator $\tilde{T}_0(s)$ is compact, and the spectral radius satisfies:

$$r(\tilde{T}_0(s)) < 1 \quad \text{for } \Re(s) > \frac{1}{2}$$

Thus, $\zeta(s) \neq 0$ in this region.

For $\Re(s) < \frac{1}{2}$, the functional equation implies $\zeta(s) = 0 \Rightarrow \zeta(1-s) = 0$, placing all zeros on the critical line. $\square$

11 Conclusion

We constructed an operator $H_\delta(s)$ whose spectral properties encode both the Euler product and the critical-line symmetry of the Riemann zeta function. The Fredholm determinant equals $\zeta(s + \delta)$, and the spectral radius transitions at $\Re(s) = \frac{1}{2}$. A final uniform Schatten convergence argument completes the proof of the Riemann Hypothesis.

References

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